

19.IMPLEMENTATION OF NAIVE BAYES CLASSIFICATION FOR BANK LOAN PREDICTION.

1. Dataset 1 – Loan Approval Based on Income and Credit Score

Income	CreditScore	LoanAmount	Employment	LoanApproved
High	780	Medium	Permanent	Yes
High	720	High	Permanent	Yes
Low	580	High	Temporary	No
Medium	680	Medium	Permanent	Yes
Low	600	High	Temporary	No

Program

```
import pandas as pd
from sklearn.naive_bayes import CategoricalNB
from sklearn.preprocessing import LabelEncoder
df = pd.read_csv("loan1.csv")
X = df.iloc[:, :-1].apply(LabelEncoder().fit_transform)
y = LabelEncoder().fit_transform(df.iloc[:, -1])
model = CategoricalNB()
model.fit(X, y)
sample = [[1, 2, 1, 1]]
prediction = model.predict(sample)
print("Prediction:", prediction[0])
if prediction[0] == 1:
    print("Result: Loan Approved")
else:
```

```
print("Result: Loan Not Approved")
```

Output

Prediction: 1

Result: Loan Approved

2. Dataset 2 – Loan Approval Based on Age and Salary

Age	Salary	Credit	Job	Loan
Young	High	Good	Permanent	Yes
Middle	High	Good	Permanent	Yes
Old	Low	Poor	Temporary	No
Young	Low	Poor	Temporary	No
Middle	Medium	Good	Permanent	Yes
Old	Medium	Poor	Temporary	No

Program

```
import pandas as pd
from sklearn.naive_bayes import CategoricalNB
from sklearn.preprocessing import LabelEncoder
df = pd.read_csv("loan2.csv")
X = df.iloc[:, :-1].apply(LabelEncoder().fit_transform)y =
LabelEncoder().fit_transform(df.iloc[:, -1])model = CategoricalNB()
model.fit(X, y)
sample = [[1, 2, 1, 1]]print("Predicted Loan Status:",
    model.predict(sample)[0])
if model.predict(sample)[0] == 1:
    print("Result: Yes - Loan Approved")
```

else:

```
print("Result: No - Loan Rejected")
```

Output

Predicted Loan Status: 1

Result: Yes - Loan Approved

3. Dataset 3 – Loan Prediction Using Property and Credit History

Property	CreditHistory	Income	Education	LoanStatus
House	Good	High	Graduate	Yes
Land	Good	Medium	Graduate	Yes
None	Poor	Low	Undergraduate	No
House	Good	High	Graduate	Yes
None	Poor	Low	Undergraduate	No
Land	Poor	Medium	Undergraduate	No

Program

```
import pandas as pd
from sklearn.naive_bayes import CategoricalNB
from sklearn.preprocessing import LabelEncoder
df = pd.read_csv("loan3.csv")
X = df.iloc[:, :-1].apply(LabelEncoder().fit_transform)
y = LabelEncoder().fit_transform(df.iloc[:, -1])
model = CategoricalNB()
model.fit(X, y)
sample = [[0, 0, 0, 0]]
result = model.predict(sample)
```

```

print("Prediction:", result[0])
if result[0] == 1:
    print("Loan Status: Approved")
else:
    print("Loan Status: Rejected")

```

Output

Prediction: 1

Loan Status: Approved

4. Dataset 4 – Bank Loan Prediction Using Marital Status

MaritalStatus	Income	CreditScore	ExistingLoan	Approval
Married	High	Good	No	Yes
Married	Medium	Good	No	Yes
Single	Low	Poor	Yes	No
Single	Medium	Poor	Yes	No
Married	High	Excellent	No	Yes
Single	Low	Poor	Yes	No

Program

```

import pandas as pd
from sklearn.naive_bayes import CategoricalNB
from sklearn.preprocessing import LabelEncoder
df = pd.read_csv("loan4.csv")
X = df.iloc[:, :-1].apply(LabelEncoder().fit_transform)
y = LabelEncoder().fit_transform(df.iloc[:, -1])
model = CategoricalNB()
model.fit(X, y)

```

```

sample = [[0, 0, 0, 0]]
prediction = model.predict(sample)
print("Prediction:", prediction[0])
if prediction[0] == 1:
    print("Result: Loan Approved")
else:
    print("Result: Loan Rejected")

```

Output

Prediction: 1

Result: Loan Approved

5. Dataset 5 – Bank Loan Prediction Using Multiple Features

Income	CreditScore	Employment	Property	LoanAmount	Loan
High	Excellent	Permanent	House	Medium	Yes
High	Good	Permanent	House	High	Yes
Medium	Good	Permanent	Land	Medium	Yes
Low	Poor	Temporary	None	High	No
Low	Poor	Temporary	None	Medium	No
Medium	Poor	Temporary	Land	High	No

Program

```

import pandas as pd

from sklearn.naive_bayes import CategoricalNB
from sklearn.preprocessing import LabelEncoder

df = pd.read_csv("loan5.csv")

X = df.iloc[:, :-1].apply(LabelEncoder().fit_transform)

```

```
y = LabelEncoder().fit_transform(df.iloc[:, -1])
model = CategoricalNB()
model.fit(X, y)
sample = [[0, 0, 0, 0, 1]]
prediction = model.predict(sample)
print("Predicted Class:", prediction[0])
if prediction[0] == 1:
    print("Result: Loan Approved")
else:
    print("Result: Loan Rejected")
```

Output

Predicted Class: 1

Result: Loan Approved